

L1.8 Linearization & Taylor

§2.9, §11.10 — what you need to know

LINEARIZATION

$$L(x) = f(a) + f'(a)(x - a)$$

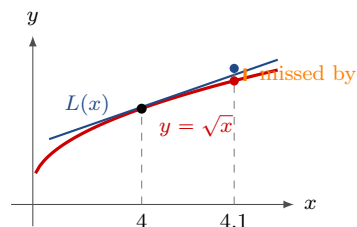
“a formula for finding the TL directly”

Ex. $f(x) = \sqrt{x}$ about $x = 4$: $f(4) = 2$, $f'(x) = \frac{1}{2\sqrt{x}}$, $f'(4) = \frac{1}{4}$

$$L(x) = 2 + \frac{1}{4}(x - 4)$$

$$\sqrt{4.1} \approx L(4.1) = 2 + \frac{1}{4}(0.1) = 2.025$$

The true value is $\sqrt{4.1} = 2.02485\dots$, so L missed by about 0.00015.



Small-angle approximations are the linearizations of $\sin x$, $\cos x$, $\tan x$ at $a = 0$:

$$\sin x \approx \sin 0 + \cos 0 \cdot x = x, \quad \cos x \approx \cos 0 - \sin 0 \cdot x = 1, \quad \tan x \approx \tan 0 + \sec^2 0 \cdot x = x$$

They are good for $x < 10^\circ$, or 0.2 radians, roughly. **Radians only!**

Δy versus dy

$$\Delta y = f(x + \Delta x) - f(x) \quad dy = f'(x) dx$$

“ Δy is the real change in the value of the function; dy is the change along the tangent line”

Take $dx = \Delta x$. The gap $\Delta y - dy$ is **how far the tangent line’s estimate missed** — halve dx and it drops by a factor of four.

Ex. $f(x) = x^2$ about $x = 3$:

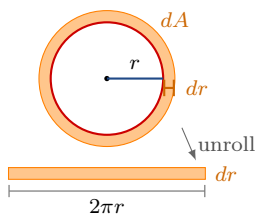
dx	Δy	$dy = 2x dx$	$\Delta y - dy$
0.1	$3.1^2 - 9 = 0.61$	0.6	0.01
0.05	$3.05^2 - 9 = 0.3025$	0.3	0.0025

UNCERTAINTY

When scientists make measurements, they measure a value and an **uncertainty**.

Ex. A scientist measures the radius of a circle to be 20 ± 0.05 cm. What is the uncertainty in its area?

The uncertainty in r is the differential $dr = 0.05$ cm, and the uncertainty in the area $A = \pi r^2$ is the differential dA : the thin ring of width dr at the edge — unrolled, a strip $2\pi r$ long and dr thick.



$$dA = A'(r) dr = 2\pi r dr = 2\pi(20)(0.05) = 2\pi \approx 6.3 \text{ cm}^2$$

$$A = 400\pi \pm 2\pi \approx 1257 \pm 6 \text{ cm}^2$$

$$\frac{dA}{A} = \frac{2\pi}{400\pi} = 0.5\% = 2 \left(\frac{dr}{r} \right)$$

SUMMATION CLOSED FORMS

$\sum_{i=1}^n a_i = a_1 + a_2 + \dots + a_n$, where i is the **index**.

$$\sum_{i=1}^n i = \frac{n(n+1)}{2} \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} \quad \sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2$$

TAYLOR AND MACLAURIN SERIES

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n \quad a=0 \Rightarrow \text{Maclaurin}$$

Ex. $f(x) = \sqrt{x}$ about $x = 4$:

n	$f^{(n)}(x)$	$f^{(n)}(4)$	$\frac{f^{(n)}(4)}{n!}$
0	$x^{1/2}$	2	2
1	$\frac{1}{2}x^{-1/2}$	$\frac{1}{4}$	$\frac{1}{4}$
2	$-\frac{1}{4}x^{-3/2}$	$-\frac{1}{32}$	$-\frac{1}{64}$

$$\sqrt{x} \approx 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2$$

$$\sqrt{4.1} \approx 2 + 0.025 - 0.00016 = 2.02484$$

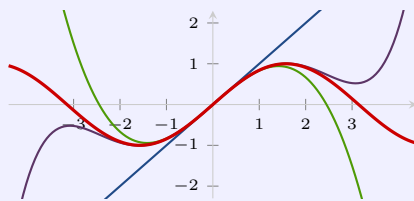
The first two terms are $L(x)$ about $x = 4$. The true value is $\sqrt{4.1} = 2.024846\dots$

Common Maclaurin series

$$e^x = 1 + x + \frac{x^2}{2!} + \dots \quad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

The truncation ladder

We can expand the Taylor series out to varying numbers of terms; the more terms we add, the better the approximation to the curve. Drawn on $\sin x$ about $x = 0$, whose even terms are all zero.



$$\begin{aligned} \sin x & \text{ the function} \\ T_1 = x & \text{ this is } L(x) \\ T_3 = x - \frac{x^3}{3!} \\ T_5 = x - \frac{x^3}{3!} + \frac{x^5}{5!} \end{aligned}$$

WATCH OUT

✗ $\Delta y = dy \rightarrow \Delta y$ is the real change in f ; dy is the change along the tangent line. They are close only when dx is small

✗ the uncertainty in A equals the uncertainty in $r \rightarrow dA = 2\pi r dr$, so the fractional uncertainty doubles: $\frac{dA}{A} = 2 \frac{dr}{r}$

✗ a Taylor polynomial is accurate everywhere $\rightarrow$ it is exact at a and accurate only *near* a ; more terms widen that interval